

Zarif Sharifovich Tadjibaev, PhD

WORLD'S FIRST **IDEAL SEWING TECHNOLOGY** **ZARIF 2025**

BASED ON A ROTATING LOOPER

**AS A BASIS FOR THE INTEGRATION
OF HUMANOID ROBOTS AND THE CREATION
OF AUTONOMOUS, UNMANNED
SEWING PRODUCTIONS OF THE FUTURE**



ZARIF 2025 PLATFORM

Ideal sewing technology ZARIF 2025 + AI + Humanoid robots + Autonomous robotic carts

PH.D. ZARIF SHARIFOVICH TADJIBAEV

**THE WORLD'S FIRST
IDEAL SEWING TECHNOLOGY ZARIF 2025
BASED ON THE ROTARY LOOPER**

**AS A FOUNDATION FOR THE INTEGRATION OF
HUMANOID ROBOTS AND THE CREATION OF
AUTONOMOUS UNMANNED SEWING PRODUCTION
FACILITIES OF THE FUTURE**

ZARIF 2025 PLATFORM

 **Ideal Sewing Technology ZARIF 2025**  **Humanoid Robots**
 **Artificial Intelligence**  **Autonomous Robo-Carts**

The Foundation of the Future Automated Sewing Factory Without Personnel

Zarif Sharifovich Tadjibaev

Ph.D. in Technical Sciences, Inventor of the Ideal Sewing Technology ZARIF 2025

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Tashkent, 2026

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ABOUT THE BOOK

This is the third book in the series **"ZARIF 2025 — The Autonomous Sewing Production Platform"**. It is a comprehensive book about how the ZARIF 2025 technology can become the machine foundation for the future autonomous sewing production, where the human role gradually transitions from manual execution of operations to task setting, engineering control, and strategic management.

The main theme of this book is not simply a new sewing machine. The main theme is the integration of four elements into a single production platform: the ideal sewing technology ZARIF 2025, digital control at the machine level, humanoid robot operators, and autonomous internal logistics. It is precisely this combination that can transform the sewing shop from the "last fortress of manual labor" into an industry compatible with artificial intelligence and round-the-clock robotic operation.

⚠ IMPORTANT NOTICE

Currently, a prototype of the ZARIF 2025 sewing machine exists, which has practically confirmed the operability of the stitch formation technology at speeds up to 5,000 stitches per minute even in very poor technical condition. Industrial machines, serial sewing automats, integrated humanoid robots, and the fully autonomous factory are currently the subject of further engineering development.

WHO IS THIS BOOK FOR

- Mechanical engineers and sewing industry technologists
- Factory managers and production directors
- Investors in industrial automation and robotics
- Physical AI and machine vision system developers
- Humanoid robot developers (Tesla, Figure, Boston Dynamics and similar)
- OEM sewing equipment manufacturers
- Digital production transformation specialists
- Everyone who wants to understand why genuine automation of sewing production requires not only a robot, but also the correct sewing technology

HOW TO READ THIS BOOK

This book fundamentally separates two levels of information.

✓ **ALREADY PROVEN BY THE ZARIF 2025 PROTOTYPE**
Operation of the stitch formation principle in one direction

Sewing without skipped stitches, thread breakage or needle breakage
Operation at speeds up to 5,000 stitches per minute
Sewing materials up to 8 mm thick with a needle bar stroke of 32 mm
Minimum stitch length of 0.5 mm
Ability to work with a standard needle with a single long groove

◆ ENGINEERING CONCEPT

Industrial version, digital control, automatic thread trimming
Circular sewing mechanism, interfaces for humanoid robots
Autonomous robo-carts and the full-scale ZARIF Autonomous Factory
These solutions are described as a realistic next stage, based on existing components, materials, and digital technologies.

AUTHOR'S PREFACE

Dear Reader!

What you hold before you is not merely a technical book, nor simply a futurological scenario. It is an attempt to honestly and consistently describe a bridge between two worlds: between modern robotics, which has already learned to walk, see, and manipulate objects, and sewing production, which still relies on the mechanical principles of the 19th century.

For almost thirty-one years, my engineering work has been connected to one main question: is it possible to build a fundamentally different technology for straight-line sewing that would simultaneously replace the weak points of the Lockstitch, eliminate the design limitations of the traditional chain stitch, and become a natural platform for the work of a humanoid robot? The answer to this question became the ZARIF 2025 technology.

I deliberately separate in the book what has already been proven by the prototype from what must be embodied in the industrial version. Such an approach is needed not for caution, but for trust. When an engineer makes a claim about the future, he is obligated to clearly indicate where the experimental fact ends and where the next stage of implementation begins.

The ZARIF 2025 technology is not just a new sewing machine. It is a new language in which robots and textile production will finally be able to speak to each other.

— Ph.D. Zarif Tadjibaev, Tashkent, 2026

INTRODUCTION

\$2.36 Trillion in Search of a Needle

Imagine an industry that annually produces goods worth more than the GDP of most countries in the world. An industry that employs 45 million people — more than the population of Canada, Australia, and Norway combined. An industry that has survived the industrial revolution, electrification, digitalization, and globalization, yet has not changed its primary tool in 170 years.

The sewing industry is worth **\$2.36 trillion dollars** a year. This is more than the global automotive or semiconductor markets. Yet, 90% of all operations here are still performed manually. Over the last 15 years, leading corporations — Adidas, Siemens, SoftWear Automation — have spent more than **\$220 million** attempting to automate sewing production. The result? Not a single successful, scalable model.

The answer you will find in this book is striking in its simplicity: the problem was not with the robots, but with the sewing machines themselves. All existing industrial sewing machines — whether lockstitch (type 301) or traditional chain stitch (type 401) — were invented in the 19th century. They are designed for human hands, for human intuition, for the human ability to feel the fabric, adjust tension, and replace the bobbin in time.

The Three Pillars of the Autonomous Factory of the Future

1. **ZARIF 2025 Technology** — the world's first robot-native sewing mechanism
2. **Humanoid Robots** (Tesla Optimus, Figure AI, Boston Dynamics) — which finally receive a tool compatible with their capabilities
3. **Artificial Intelligence and Digital Logistics** — machine vision, predictive maintenance, decentralized auto-carts

This book is not science fiction and not a marketing prospectus. It is an engineering chronicle of a 31-year journey from an idea to a working prototype, and simultaneously — a technical plan for transforming this prototype into an industrial revolution.